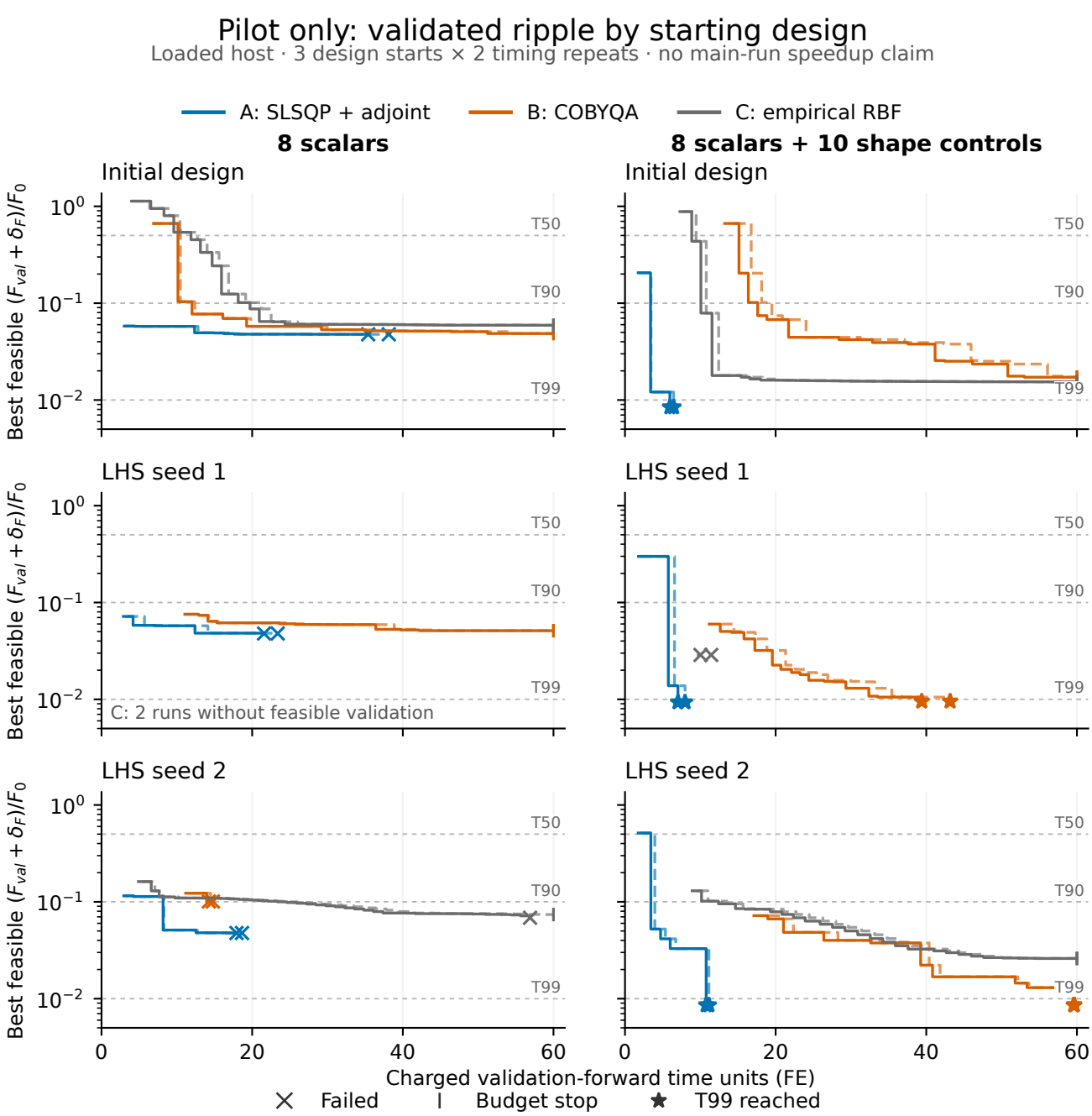
Pilot only: validated ripple by starting design
Loaded host · 3 design starts × 2 timing repeats · no main-run speedup claim
A: SLSQP + adjoint
B: COBYQA
C: empirical RBF
8 scalars
8 scalars + 10 shape controls
Initial design
LHS seed 1
LHS seed 2
Best feasible $(F_{val} + \delta_F)/F_0$
10^0
10^{-1}
10^{-2}
T50
T90
T99
C: 2 runs without feasible validation
0
20
40
60
Charged validation-forward time units (FE)
Failed
Budget stop
T99 reached
Solid / dashed = timing repeats 1 / 2, not independent designs.
Steps change only after completed feasible validations; each curve ends at its actual stop.